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(12) United States Design Patent

Sasaki et al.

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(54) **OPTICAL FIBER CONNECTOR**

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(**) Term: **15 Years**

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Related U.S. Application Data

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(30) **Foreign Application Priority Data**

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Feb. 24, 2022 (JP) .... 2022-003593 D

(51) **LOC (15) Cl.** **13-03**
(52) **U.S. Cl.**
USPC **D13/133**
(58) **Field of Classification Search**
USPC D6/682.6; D8/98, 349; D10/70, 109.1, D10/109.2, 46, 66, 81; D13/133, 147,
(Continued)

(56) **References Cited**

U.S. PATENT DOCUMENTS

5,016,968 A 5/1991 Hammond et al.
5,838,856 A 11/1998 Lee
(Continued)

OTHER PUBLICATIONS

FSG MT ferrule, posted Jun. 18, 2020 [online], [retrieved May 2, 2025]. Retrieved from internet, https://www.fsgnetworks.com/products/mt-ferrule/ (Year: 2020).*
(Continued)

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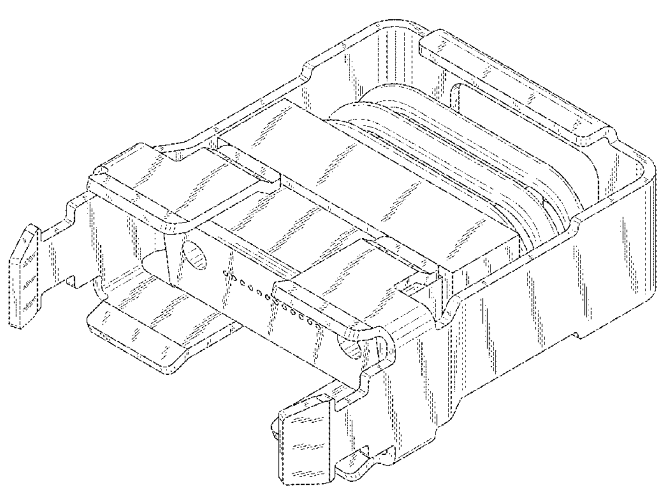
(57) **CLAIM**

The ornamental design for an optical fiber connector as shown and described.

DESCRIPTION

FIG. **1** is a front view of an optical fiber connector of our new design;
FIG. **2** is a rear view of the optical fiber connector of FIG. **1**;
FIG. **3** is a top plan view of the optical fiber connector of FIG. **1**;
FIG. **4** is a bottom plan view of the optical fiber connector of FIG. **1**;
FIG. **5** is a right side view of the optical fiber connector of FIG. **1**;
FIG. **6** is a left side view of the optical fiber connector of FIG. **1**;
FIG. **7** is a front, top plan and left side perspective view of the optical fiber connector of FIG. **1**;
FIG. **8** is a rear, bottom plan and right side perspective view of the optical fiber connector of FIG. **1**;
FIG. **9** is a sectional view taken along the line **9-9** of the optical fiber connector of FIG. **6**; and,
FIG. **10** is an environmental perspective view showing the optical fiber connector of FIG. **1**.
The features shown in evenly-dashed broken lines depict environmental subject matter only and form no part of the claimed design.

1 Claim, 10 Drawing Sheets



(58) **Field of Classification Search**

USPC D13/154, 156, 146, 101, 118, 123; D14/433, 420, 256, 428, 206, 372, 426, D14/429, 435, 453, 240, 242, 357, 358, D14/140–140, 155, 125, 137, 139, 243, D14/348, 349, 351, 354, 355, 251, 253, D14/15, 10, 122, 144; D16/134, 130, D16/131, 132, 135, 136, 202–208, 210, D16/245, 219, 221, 235, 237, 313, 314, D16/326; D22/109, 108; D23/40, 43, D23/260; D24/120, 113, 118, 119, 121, D24/122, 124, 125, 128, 137, 186, 172, D24/129; D26/24, 28, 122, 113, 118; D32/35

CPC A45C 11/182; A45C 13/002; A45F 5/00; H04M 1/21; H04M 1/04; F16B 47/003; F16M 13/04; H01R 13/443; H01R 33/965; H01R 13/52; H01R 2201/16; Y10T 29/53235; G02B 6/38875; G02B 6/3888; G02B 6/3825; G02B 6/3893; G02B 6/406; G02B 6/3895; G02B 6/387; G02B 27/026; G02B 23/16; G02B 2003/0093; G02B 3/04; G02B 6/3831; G02B 6/3887; G02B 6/3866; G02B 6/3871; G02B 6/3898; G02B 6/3869; G02B 6/3882; G02B 6/3885; G02B 6/3846; G02B 6/3802; G02B 6/3865; G02B 6/381; G02B 6/3644; G02B 6/32; G01C 3/00; G01C 3/02; G01C 3/04; G01C 3/06; G01C 3/08; G01C 3/085; G01C 3/10; G01C 3/12; G01C 3/16; G01C 3/18; G01C 3/20; G01C 3/22; G01C 3/24; G01C 3/26; G01C 3/28; G01C 3/30; G01C 3/32; B60Q 1/04; B60Q 1/26; F21S 8/026; F21S 8/04; F21V 21/02; F21V 21/04; H04J 11/00; H04J 13/00; E06B 9/24; G03B 17/02; G03B 11/00; G03B 15/14; C03B 2215/46; H04L 41/0896; H04L 41/5022; H04L 67/06; H04L 67/1097; H04L 67/34; A61B 2090/306; A61B 90/35

See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

6,530,696	B1	3/2003	Ueda et al.	
6,616,343	B2 *	9/2003	Katsura	C08K 7/08 385/60
D590,344	S *	4/2009	Suematsu	D13/147
9,465,170	B1	10/2016	Childers et al.	
9,563,027	B2	2/2017	Childers et al.	
9,817,194	B2	11/2017	Childers et al.	
D828,811	S	9/2018	Aoshima et al.	
D828,812	S	9/2018	Aoshima et al.	
11,280,966	B2	3/2022	Cloud et al.	
11,719,892	B2	8/2023	Higley et al.	
D1,038,038	S *	8/2024	Sasaki	D13/147
2002/0149126	A1 *	10/2002	Kihara	B29C 45/40 264/1.25
2013/0183004	A1	7/2013	Hughes et al.	
2013/0266271	A1	10/2013	Li et al.	
2016/0139339	A1 *	5/2016	Sasaki	G02B 6/3825 385/59
2017/0176694	A1	6/2017	Childers et al.	
2019/0361177	A1	11/2019	Aoshima et al.	
2021/0215887	A1	7/2021	Cloud et al.	
2021/0325614	A1	10/2021	Childers et al.	
2022/0026645	A1	1/2022	Arao	
2023/0036226	A1 *	2/2023	Wang	G02B 6/3882

OTHER PUBLICATIONS

US Conec MT ferrule, posted Jan. 15, 2015 [online], [retrieved May 2, 2025]. Retrieved from internet, https://www.lightwaveonline.com/directory/components/connector-components/press-release/14059062/us-conec-ltd-us-conecs-new-singlemode-pre-angled-mt-ferrule-delivers-faster-process-time-andenhanced-performanc (Year: 2015).*

FS MTP connectors, posted Sep. 27, 2024 [online], [retrieved May 2, 2025]. Retrieved from internet, https://www.fs.com/blog/a-comprehensive-guide-to-mtp-connector-958.html (Year: 2024).*

Fujikura TMT ferrule, posted Mar. 3, 2024 [online], [retrieved May 2, 2025]. Retrieved from internet, https://www.optic-product.fujikura.com/optical-components/en/products/tmt-ferrule-technology/ (Year: 2024).*

Glenair MT elite ferrule kits, posted May 24, 2024 [online], [retrieved May 2, 2025]. Retrieved from internet, https://cdn.glenair.com/fiberoptics/mt-ferrule/pdf/181-133.pdf (Year: 2024).*

Optico MT-FA fiber jumper, posted Nov. 17, 2022 [online], [retrieved May 2, 2025]. Retrieved from internet, https://www.fiberopticom.com/patch-cord/mt-patch-cord/mt-fa-fiber-jumper.html (Year: 2022).*

Sumitomo nanoplug mt, posted Aug. 10, 2022 [online], [retrieved May 2, 2025]. Retrieved from internet, https://sumitomoelectric.com/products/NanoPlug (Year: 2022).*

"Revolutional Style of MT/MPO for Single Mode", 367 Sumitomo1 NPL, posted Oct. 16, 2021 [online], [retrieved Feb. 1, 24], Retrieved from internet, https://sumitomoelectric.com/Advanced_Multi-fiber_Connectors/AirMT (Year:2021).

Feb. 15, 2024 Office Action issued in U.S. Appl. No. 29/849,367.

367 abd 375 Nanoplug NPL, posted Apr. 15, 2021 [online], [retrieved Feb. 1, 2024]. Retrieved from internet, https://sumitomoelectric.com/products/NanoPlug (Year: 2021).

May 13, 2025 Notice of Allowance issued in U.S. Appl. No. 29/938,377.

* cited by examiner

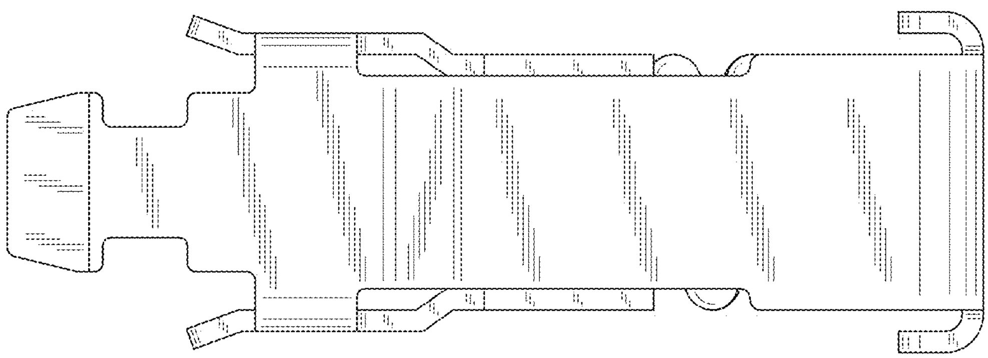


FIG. 1

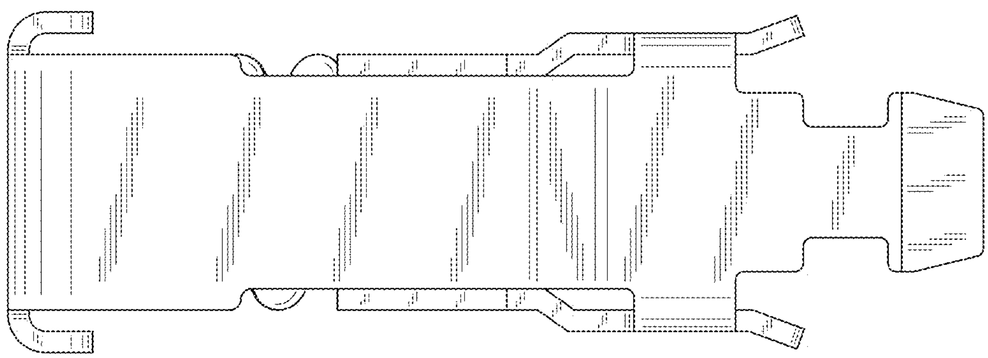


FIG. 2

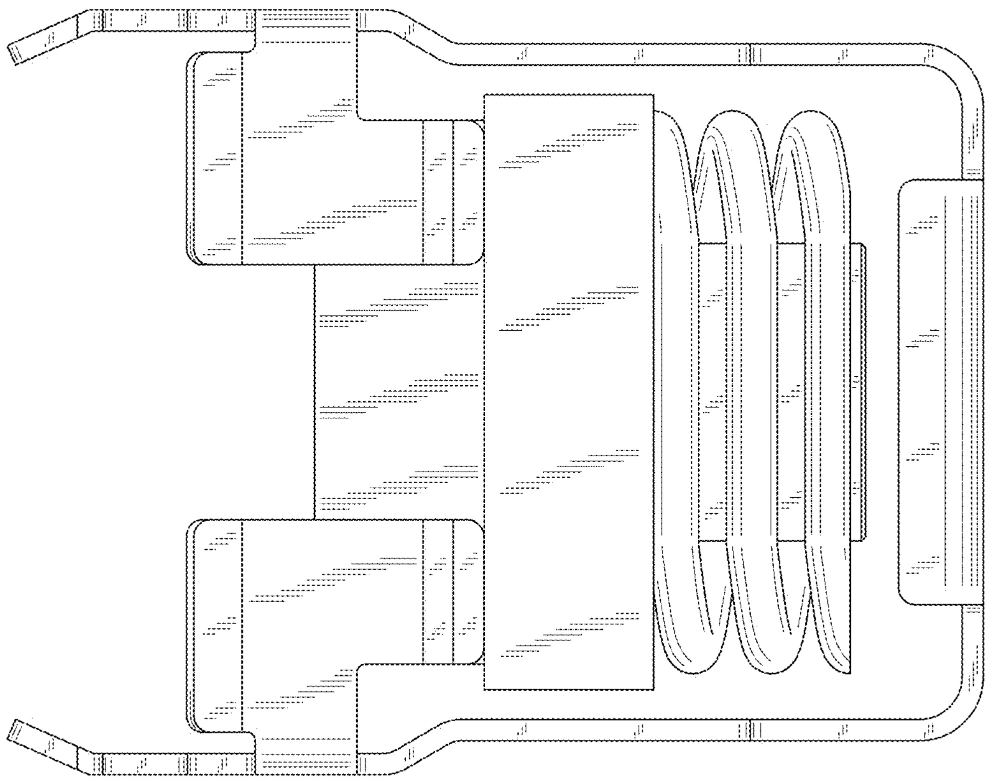


FIG. 3

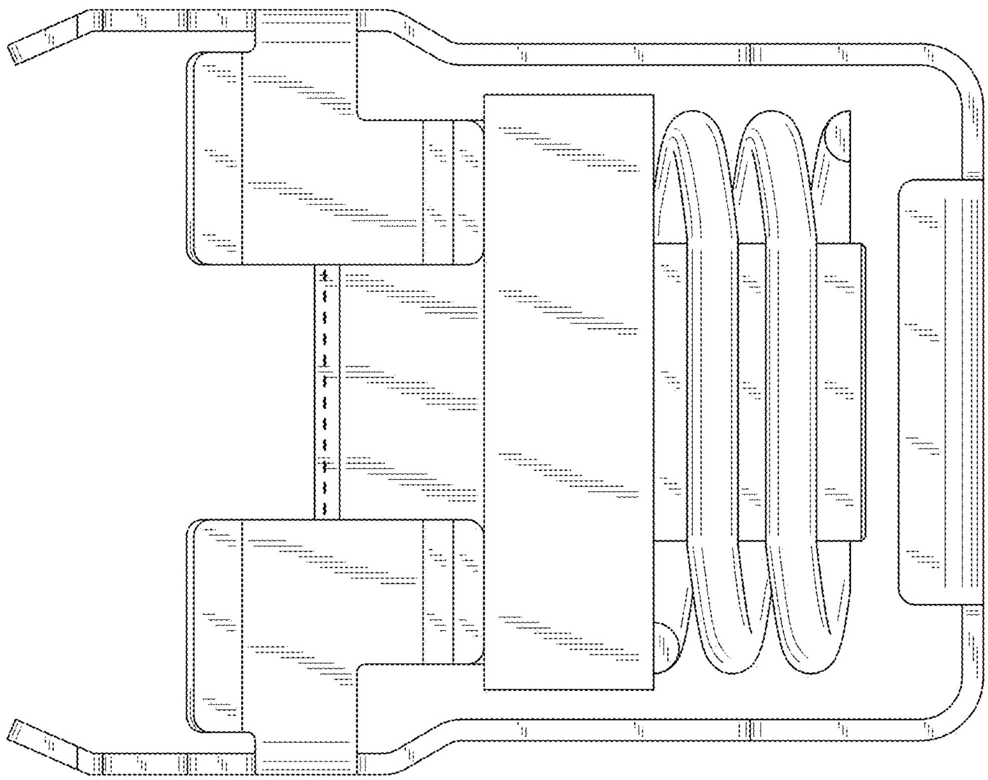


FIG. 4

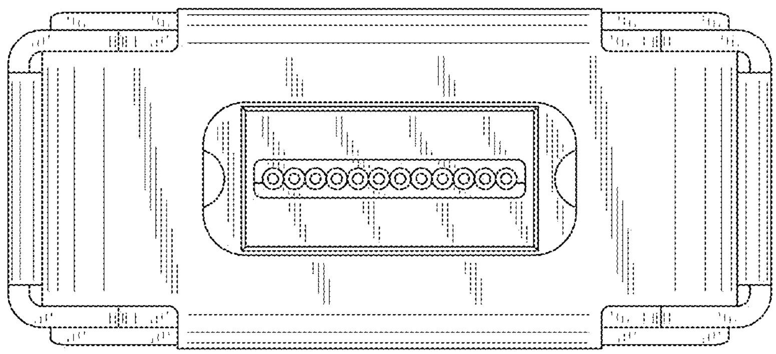


FIG. 5

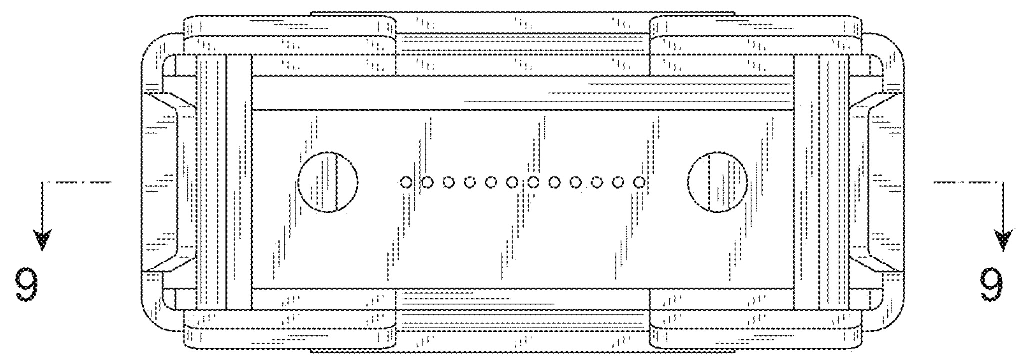


FIG. 6

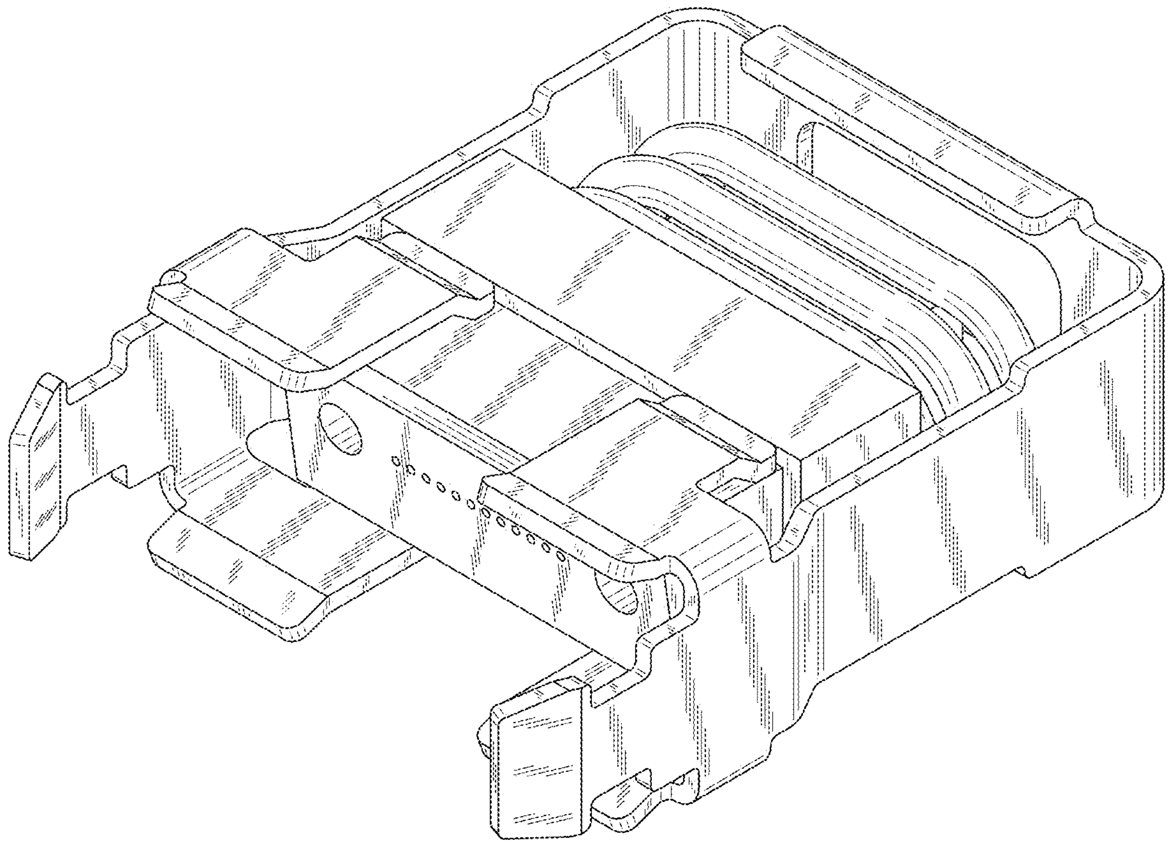


FIG. 7

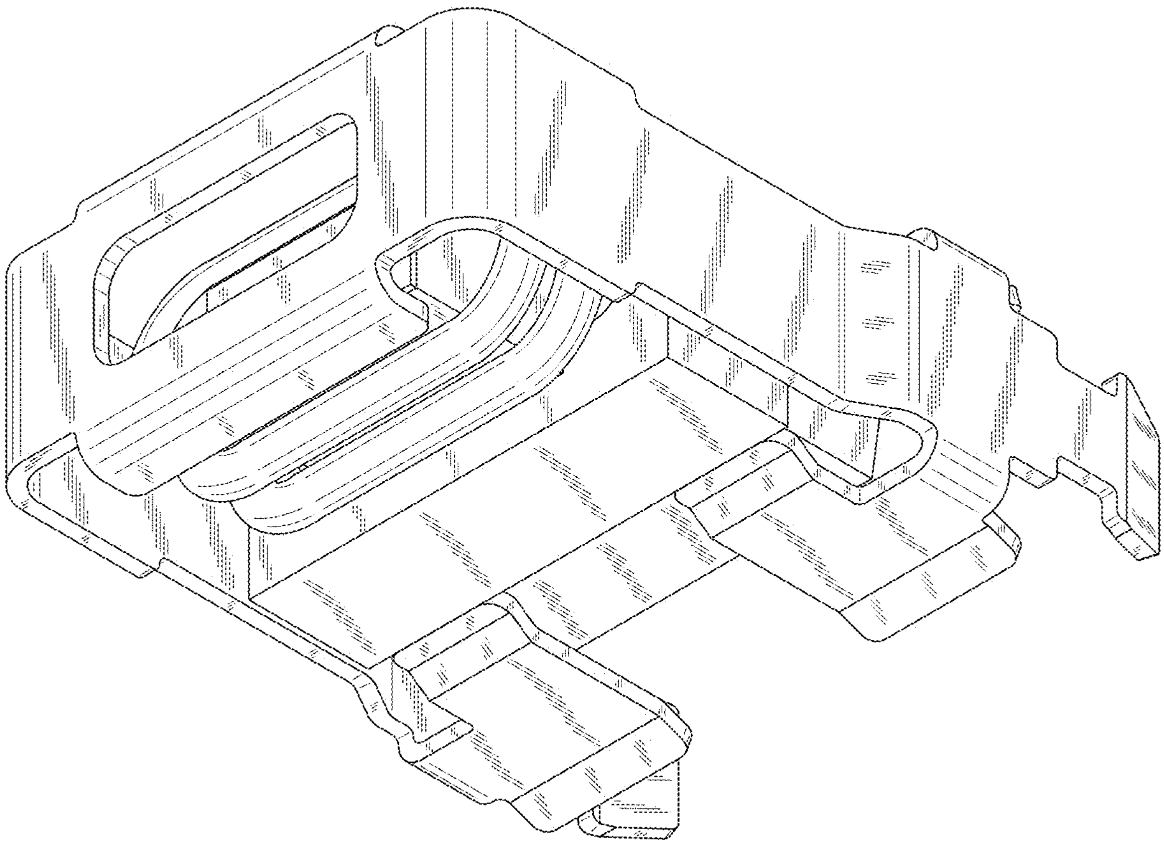


FIG. 8

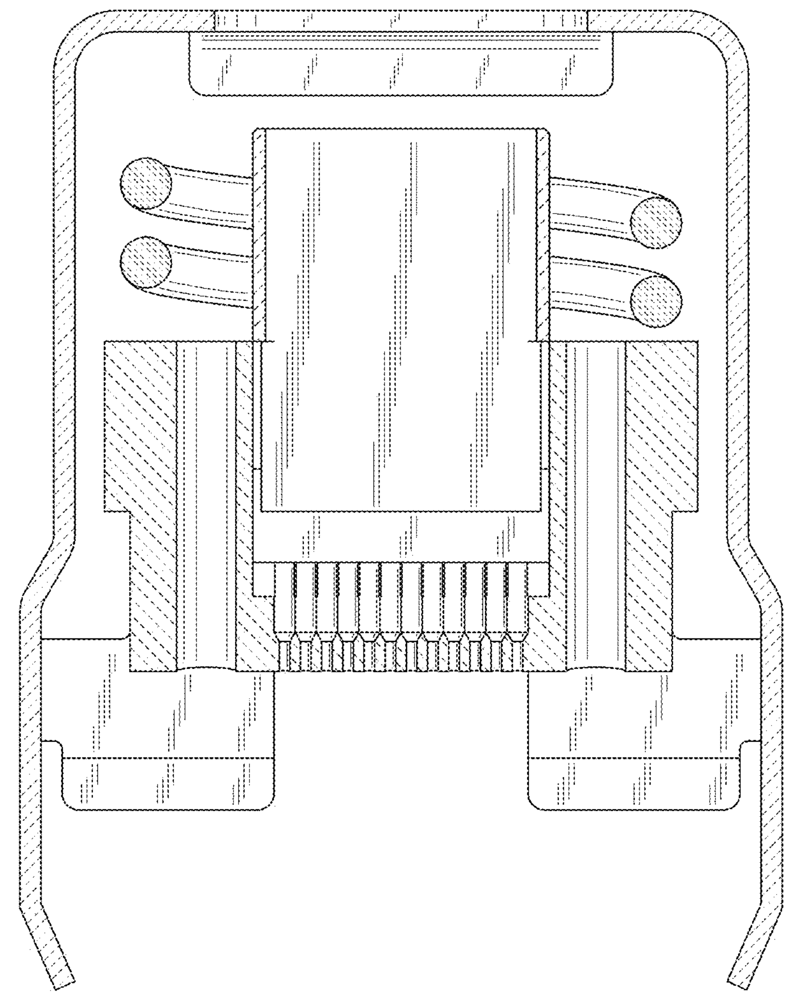


FIG. 9

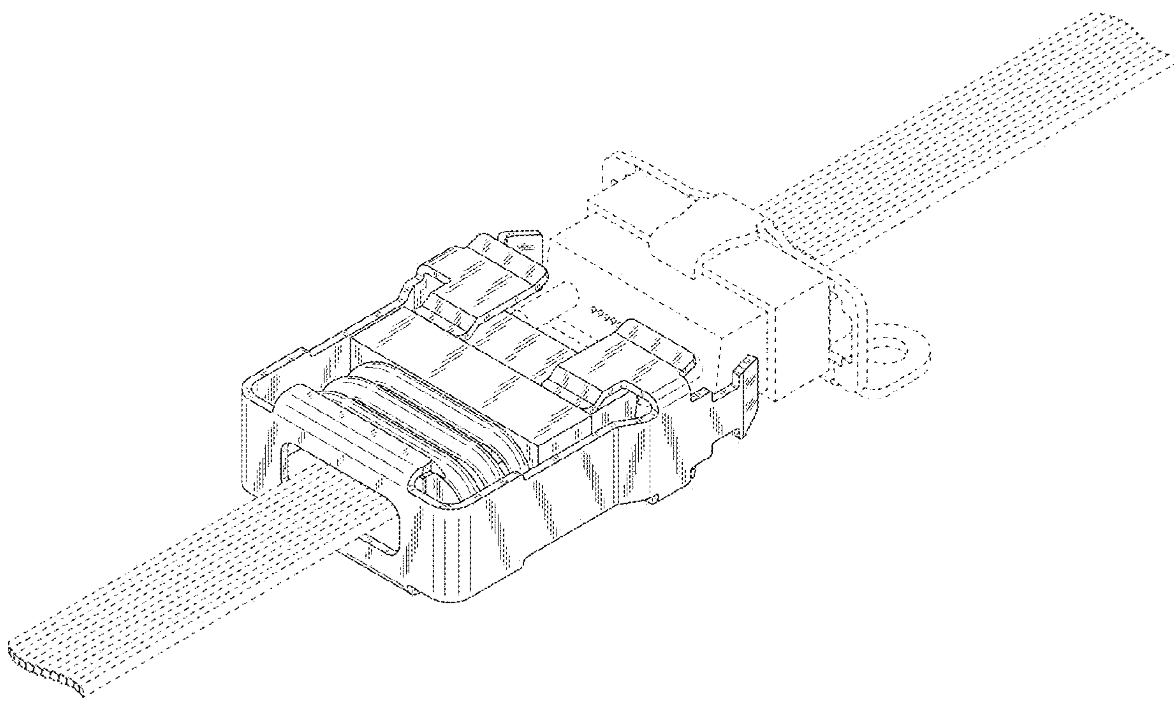


FIG. 10